

Module 7: Lesson 4

Reinforcement Learning: Optimal Trading Behavior



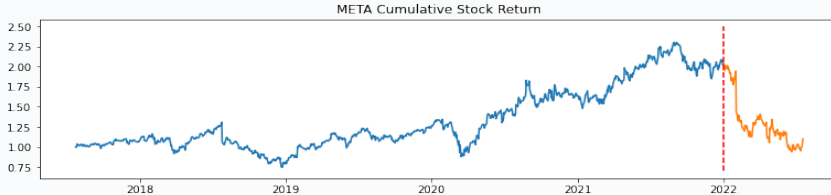
Outline

- ▶ An investor trading problem
- ▶ Finding the optimal trading choices with Q -Learning

The investor problem: Data

We are going to apply Q-learning to model an investor's optimal daily trading choices in the context of a single stock, in this case META, formerly Facebook (FB).

Our objective with this example is to train an RL algorithm during a training sample and then test its performance out-of-sample.



We select the end of 2021 as the end of the training sample.

The modeling setup

An agent (the investor) must take actions (trading decisions) based on a wealth-maximization objective with risk aversion considerations.

The agent starts with some initial cash c_0 that can be allocated to purchasing stock of META during some time period.

Terminal conditions:

- ▶ The agent reaches the terminal date T that determines the end of the training sample.
- ▶ The agent runs out of cash, in which case they incur a penalty.
- ▶ The agent reaches an upper bound or a lower bound in their returns.

In all terminal states, the agent obtains a reward that is increasing in its wealth but that takes into account risk aversion.

Trading involves some transaction costs that are deducted from the investor's wealth.

Q-Learning algorithm

We opt for a Q-Learning algorithm that updates the state-action values as:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$

where the state is represented by the current stock price of META, its current cash holdings, and the current holdings of META that the investor has.

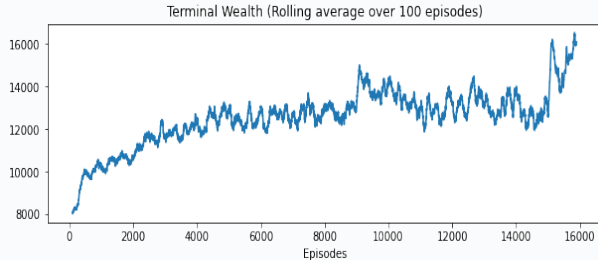
The agent's actions are the number of shares of META to hold until the next trading day.

The spaces of stock prices and cash held are continuous; we opt for the simplest nearest-neighbor interpolation to approximate the functions.

We set to 16,000 the number of episodes and the updating parameter α to 0.4.

We split the episodes into three samples that differ in the randomness of ε -greedy policies, with $\varepsilon = [0.1, 0.05, 0]$.

Q-Learning algorithm: Learning over episodes

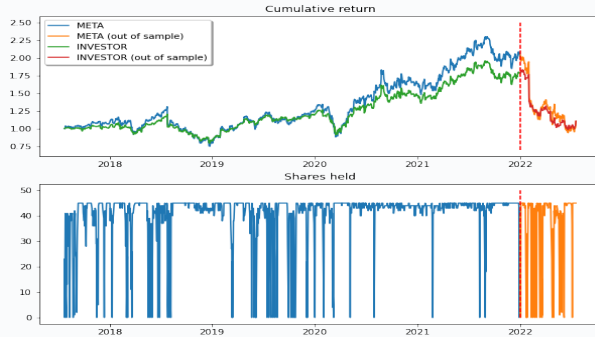


The model is increasingly able to learn about courses of action that yield the greatest wealth levels for the agent.

However, notice that maximizing wealth is not the only objective of the agent: the riskiness of the path must be taken into account.

Q-Learning algorithm: Out-of-sample performance

We simulate the performance of the model for an investor that starts with \$10,000 to invest in META and check the evolution of their wealth in the training sample and out-of-sample.



Summary of Lesson 4

In Lesson 4, we have looked at:

- ▶ Developing a model of an investor's trading choices
- ▶ Estimating the model using Q-Learning.

⇒ **References for this lesson:**

Sutton, Richard S., and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018. (see Chapter 6)

TO DO NEXT: Now, please go to the associated Jupyter Notebook for this lesson to explore the programming implementation of the investor problem described in these slides.

With this lesson, we have finished the study of the Reinforcement Learning methodologies. Feel free to enhance your knowledge by exploring more advanced methods in this exciting and growing field of Machine Learning.